

Dynamic Connectivity

Time limit: 1.00 s

Memory limit: 512 MB

Consider an undirected graph that consists of n nodes and m edges. There are two types of events that can happen:

1. A new edge is created between nodes a and b .
2. An existing edge between nodes a and b is removed.

Your task is to report the number of components after every event.

Input

The first input line has three integers n , m and k : the number of nodes, edges and events.

After this there are m lines describing the edges. Each line has two integers a and b : there is an edge between nodes a and b . There is at most one edge between any pair of nodes.

Then there are k lines describing the events. Each line has the form " $t a b$ " where t is 1 (create a new edge) or 2 (remove an edge). A new edge is always created between two nodes that do not already have an edge between them, and only existing edges can get removed.

Output

Print $k + 1$ integers: first the number of components before the first event, and after this the new number of components after each event.

Constraints

- $2 \leq n \leq 10^5$
- $1 \leq m, k \leq 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 3 3 1 4 2 3 3 5 1 2 5 2 3 5 1 1 2	2 2 2 1